Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/01/2018
PI Name:	Monshausen, Gabriele Ph.D.		
Project Title:	Defining the Role of the Receptor Like Kinase FERONIA in Plant Architecture Development Under Mechanical Load		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-5301	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date change to 7/01/2018 per NSSC information (Ed., 11/22/17) NOTE: End date change to 11/1/2017 per NSSC information (Ed., 4/5/16) NOTE: Period of performance corrected to actual dates per NSSC information (Ed., 9/10/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	The shaping of plant architecture is a process that unfolds over the entire lifespan of a plant and is continually informed by developmental and environmental cues. On Earth, the most pervasive and constant of these environmental cues is gravity. Gravity is an important factor in guiding how plant organs are organized in space. However, gravity does not only provide directional information during plant development. As growing plants gain size and mass, the increasing weight of vertical stems and laterally projecting branches and leaves results in progressive mechanical self-loading. These mechanical loads can lead to buckling or breaking of plant organs unless they are structurally reinforced. The principal objective of this research proposal is to understand how mechanical loads are translated into developmental programs that maintain structural integrity of the plant body. Specifically, we will investigate the role of the plasma membrane receptor-like kinase FERONIA in this process, as we have recently discovered that FERONIA is crucial to normal mechanical signaling and mechanical development of the model plant Arabidopsis thaliana (Mouse Ear Cress). We will employ fluorescence microscopy, growth kinematics, and molecular biology approaches to investigate how FERONIA interacts with gravity and other mechanical signal transduction pathways. Results from such analyses are responsive to the goals of the NASA Research Announcement in addressing the fundamental question of how plants sense and react to gravity as a mechanical force to regulate growth and development. Understanding how FERONIA serves to help plants adapt under conditions of non-Earth-normal gravity may eventually provide a means for modulating mechanical strength in plants raised in microgravity conditions in the absence of mechanical self-load.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research addresses the question of how plants sense mechanical forces and how they utilize this information to modulate growth and tissue strength. Understanding how mechanical signaling, and specifically mechanical signaling through the receptor-like kinase FERONIA, affects the growth and development of plants will not only provide insight into how the reduced gravity environment of spaceflight will affect plant architecture, but will also have translational potential for improving plant agriculture on Earth. Understanding the mechanisms that enhance plant tissue strength may eventually yield molecular targets for the manipulation of traits that allow plants to withstand wind and to penetrate compacted soils.
Task Progress:	The receptor-like kinase FERONIA is activated by binding its small secreted peptide ligand RALF1 (RAPID ALKALINIZATION FACTOR1) but is also essential to mechanical signal transduction. To understand the role of FERONIA in mechanical signal transduction we have analyzed (1) the effect of RALF1 on mechanosensitivity of Arabidopsis roots; (2) the effects of RALF1 and mechanical wounding on FERONIA subcellular localization; (3) mechanisms underlying FERONIA-dependent Ca2+ signaling. We have found that pretreatment with RALF1 peptides rapidly desensitizes FER-dependent mechanical Ca2+ signaling. We further observed that RALF1 triggers endocytosis and degradation of FERONIA via the endosomal pathway. In contrast, mechanical wounding also appears to elicit a loss of FERONIA from the plasma membrane, but via a different pathway. Finally, we found that FERONIA-dependent mechanical Ca2+ signaling requires the expression of a subset of CYCLIC NUCLEOTIDE-GATED CHANNELS; the same channels are not essential for FERONIA-dependent RALF1-induced Ca2+ signaling. These results suggests that FERONIA may act as a dual function receptor or as a hub where RALF1 and mechanical signaling pathways (among others) are integrated. To understand the role of FERONIA in regulating cell expansion, we have (1) assayed rapid growth responses to RALF1 using high-resolution kinematic analysis ; (2) analyzed FER-dependent changes in vacuolar morphology and cytoskeletal organization. We found that RALF1 triggers rapid growth inhibition in Arabidopsis roots in a FERONIA- and Ca2+ dependent manner. RALF1-induced growth inhibition is correlated with a reduction in cell extensibility as assayed in osmotic inflation experiments. Osmotic inflation assays also indicate differences in wild type and feronia loss-of-function mutant cell extensibility. We further observed that RALF1 treatment triggers a rapid (within 10 minutes) change in vacuolar morphology that is associated with a concomitant change in actin cytoskeleton organization. These results are consistent with FERONIA signaling affeccting both cell wall extensibility and cell turgor to modulate growth.
Bibliography Type:	Description: (Last Updated: 12/06/2016)